Gesture Controlled Virtual Mouse

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Abstract - A gesture-controlled virtual mouse provides an intuitive, hands-free way to interact with computers, addressing ergonomic challenges, accessibility issues, and the need to reduce physical contact in certain environments, such as medical settings. Traditional systems that rely on external devices, such as sensor gloves, or outdated machine learning methods such as SVM, face limitations such as imprecise gesture recognition, latency, and the need for calibration.The proposed system uses MediaPipe's machine learning framework with CNN to enable realtime hand detection and gesture control via a standard web camera, eliminating the need for external hardware. It allows users to control the cursor, click, scroll and adjust brightness with finger gestures, providing a smooth, responsive and accessible solution for various applications.

Keywords – Gesture-Controlled Virtual Mouse, Hands- free Interaction, Media Pipe Framework, CNN, Hand Gestures.

I. INTRODUCTION

The paper on the development of a gesture-controlled virtual mouse system is presented and how users can control their computer's cursor using intuitive hand movements. It uses computer vision and machine learning technologies, making it an easy, hands-free alternative to a traditional input device such as a mouse or keyboard. This can help increase accessibility for individuals with some form of physical disability while providing a hygienic touch-free solution in healthcare or otherwise where contact should be minimized.

Quite a long way from where it came, gesture control has developed in stages from early research work starting even in the 1970s to the 1990s. The very beginning of this development was made by Douglas Engelbart in the 1960s with the development of the first traditional computer mouse and the popularisation of GUIs which made the mouse an indispensable input device. In the late 1980s to the 1990s, touchpads were the alternative way of controlling the cursor, especially for laptops. It is only in the late 1990s to early 2000s that gesture recognition research started to take off, looking into possibilities of using camera-captured hand movements to control a computer. Such systems in these early days were plagued by accuracy and practicality issues.

Alternative input devices like stylus pens, and motion-sensing peripherals in the Nintendo Wii were also introduced in the 2000s that facilitated new virtual mouse concepts. The computer vision of the 2010s was quite advanced through the spread of smartphones, touch screens, and gesture-based interactions. Computer vision frameworks such as OpenCV and MediaPipe by mid2010s sprang up allowing complex gesture recognition with regular webcams running in real-time.

Recent developments have further consolidated the gesture-controlled virtual mouse system, making control real-time and responsive without the use of any hardware or any significant calibration. Today's systems, as depicted in this work of research, support several gestures including cursor control, clicks, scrolling, and changing brightness by single and multi-finger movements. The system is also touchless in nature, thereby not only reducing physical strain but also hygienic because it bans the sharing of input devices, especially for health care settings or presentations. The focus of the system on accessibility enables mobility-impaired users to enjoy the advantages of more ergonomic and comfortable interface options. This means that the proposed system is using only a standard webcam, for which there is no need to purchase special hardware and through the use of this system, the broad scope of applications was covered in a simple and costfree manner.

The rest of the paper is structured as follows: Section 2 provides a comprehensive literature review, discussing existing technologies and methodologies related to gesture recognition. Section 3 explains in detail about the technologies used in our system. In Section 4, we present the results of our implementation, supported by screenshots and detailed analysis. Finally, the article concludes with the conclusion and suggestions for future work in this emerging field.

II. LITERATURE SURVEY

By using hand gestures instead of a physical device, the suggested solution allows users to control their computer mouse. In order to classify these motions according to a pre-trained model, it uses Convolutional Neural Networks (CNNs) after using MediaPipe for gesture identification through webcams. Control is made intuitive by converting recognized motions into cursor actions. Real-time interaction is guaranteed by the system, which provides fast cursor updates and gesture interpretation. OpenCV improves the webcam stream to increase the accuracy of gesture recognition, giving consumers a comfortable and easy way to utilize hand gestures to interact with computers.

Using hand gesture detection to implement a virtual mouse system requires multiple phases. The first step is to utilize a camera to take pictures of the user's hand in order to recognize gestures in real time. Convolutional neural networks (CNN), which are trained on a variety of hand movement data to ensure accuracy, are the specific type of artificial intelligence that the system uses. Mouse operations, such as clicks or movements, are converted from detected gestures on a virtual screen. For people who have trouble using conventional devices, the design uses computer vision and artificial intelligence to develop an intuitive interface that is scalable and can accommodate a variety of hand movements and user needs.

This study describes hand gesture recognition, a

promising technique that offers users easy and natural ways to interact with digital interfaces. The suggested system uses computer vision techniques to track and identify hand movements in real time. By linking particular motions to mouse control commands, users can explore and engage with graphical user interfaces (GUIs) without the need for physical input devices. This article describes a hand gesture-controlled virtual mouse system that makes use of artificial intelligence and machine learning.

In this paper, an AI-powered hand gesture recognition system for a virtual mouse is proposed. Constructed for users who have trouble using conventional input devices, the system uses a camera to take pictures of hands, interprets them using computer vision and machine learning methods, and converts identified motions into mouse clicks on a virtual screen. Voice commands and dynamic/static gestures are supported, and no more hardware is needed. The system may be used in dangerous situations and improves user accessibility in humancomputer interaction. It is also scalable and flexible.

A webcam is used for input, and the system is an AIdriven interface that uses hand gestures to control a mouse and keyboard and displays results on the screen. It improves intelligent learning systems and is made to be used in pandemic situations. It is implemented using Python and OpenCV. The suggested solution uses AI techniques to improve the gesture-controlled virtual mouse and virtual keyboard through a virtual assistant.

The suggested solution in this work uses an ordinary webcam and Python with OpenCV to provide a transparent interface that allows people to communicate with their computers. It seeks to lessen reliance on physical input devices and minimize touch with peripherals by converting hand movements into mouse operations. The development must overcome obstacles like preserving real-time speed and guaranteeing robustness under a variety of circumstances. With the help of this system, which converts hand gestures into mouse motions, the user may interact with apps, traverse the graphical user interface (GUI), and carry out a variety of tasks using natural hand movements.

Without the use of conventional input devices like a mouse or keyboard, the suggested method enables users to create and navigate presentation slides using hand gestures. It uses hand landmark recognition to digitize the blackboard-chalk method and recognize five distinct gestures: using the thumb to move to the next slide, the little finger to go back to the previous slide, the two fingers to display a pointer, and the one finger to draw. This invention meets the growing demand for portable electronics in classrooms.

This project investigated the use of fingertip recognition to control mouse clicks and motions via hand gestures. It is divided into three primary stages: mouse control, hand motion recognition, and color-based finger detection. Its major objective is to track fingers using colored caps. In order to provide precise cursor control in the system, users can efficiently execute movements by affixing RGB color tapes to their fingertips.

In order to eliminate the need for actual devices like wireless mice and dongles, the suggested virtual mouse and keyboard system records hand movements and colored objects using a webcam or built-in camera. Convolutional Neural Networks (CNNs) are used to identify hand motions, allowing users to virtually use the computer's cursor functions, such as scrolling and clicking. It also uses image processing techniques to operate the virtual keyboard. By enabling virtual presentations, which is particularly helpful in settings like conferences, this technology lowers the possibility of COVID-19 transmission.

This research suggests a contactless system that improves human-computer interaction by using hand gestures in place of conventional input devices like a mouse. It aims to offer a more practical and hygienic way to interact with interfaces, and it may find usage in public interfaces, gaming, and healthcare. Since there are currently few gesture-controlled gadgets available, our suggested solution has dynamic gesture recognition and robust hand tracking for precise operation. In order to execute a variety of mouse activities, the system efficiently recognizes and interprets hand motions utilizing the Mediapipe architecture and computer vision algorithms. Performance measurements and system development are discussed in the essay.

III. METHODOLOGY

The suggested solution uses a regular webcam to record live video and MediaPipe to track and detect hand gestures. Important elements include gathering data, using machine learning to recognize gestures, and integrating libraries like PyAutoGUI to carry out mouse operations. This methodical methodology guarantees accurate recognition and reaction, providing a smooth user experience. The development and implementation process will be thoroughly outlined in the parts that follow.

Convolutional Neural Network (CNN)

Convolutional Neural Networks (CNNs), the most popular deep learning technology, are designed for structured grid data, like images. Convolutional layers with filters that automatically learn spatial hierarchies of features are used to mimic human visual perception. With a sequence of convolutional and pooling layers, CNNs effectively lower spatial dimensions while capturing ever-more-complex information. Each filter identifies unique patterns. CNNs are essential for computer vision applications since they are skilled at tasks like object detection, gesture recognition, and image classification because they can recognize patterns in visual input without the requirement for manual feature extraction.

Media Pipe

An open-source framework for building multimodal applied machine learning pipelines, MediaPipe was developed by Google. For tasks like gesture recognition, hand tracking, object detection, and facial recognition, it provides a vast array of pre-built models and solutions. A virtual mouse system that uses hand gestures must have MediaPipe's hand tracking module. It recognizes and tracks the locations of significant hand landmarks in real time using advanced machine learning techniques, particularly Convolutional Neural Networks (CNNs). This allows the system to accurately read user gestures, including finger movements and attitudes, facilitating seamless interaction without the need for physical input devices. Because of its simplicity of integration and realtime processing capabilities, MediaPipe is an ideal choice for developing gesture-based applications.

PyautoGUI

An operating system's graphical user interface (GUI) can be interactd with automatically using the cross-platform Python module PyAutoGUI. It gives programmers a userfriendly API that lets them programmatically operate the keyboard and mouse, opening up a world of automated tasks. Users may move the mouse pointer, click, drag, scroll, type text, and take screenshots with PyAutoGUI, making it an essential tool for automating repetitive tasks or creating original interaction models. With the use of system gestures, it translates hand movements into corresponding mouse actions. This seamless connection allows users to use laptops hands-free with simple gestures. The cross-platform compatibility of PyAutoGUI guarantees that it can function on a range of operating systems without requiring significant modifications, which enhances the project's usefulness and accessibility.

Architecture Diagram

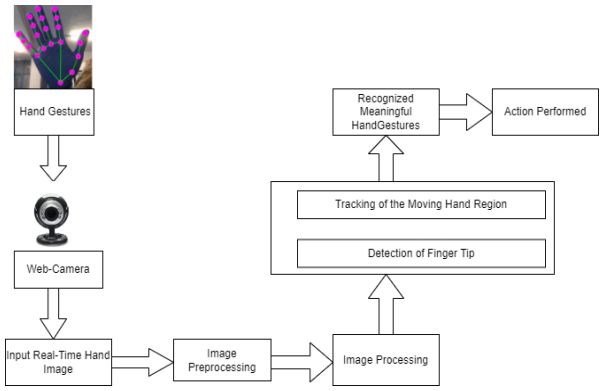

Fig:1

The hand gesture-controlled virtual mouse technology employs a methodical technique to enable people to exchange natural movements with their computers. A webcam captures users' hand movements in real time, which are subsequently preprocessed to reduce noise and improve clarity. To detect important gestures, the system then follows the moving hand's region and searches for certain finger tips. Scrolling, clicking, and pointer motions are examples of actions that are translated to correspond with these gestures. This architecture eliminates the need for traditional input devices and facilitates seamless human-computer interaction, improving accessibility and user experience.

Technologies

Python

Python is a flexible programming language that may be used by novices and experts alike in a variety of domains, including computer vision, data science, machine learning, web development, and automation. It integrates libraries like OpenCV, MediaPipe, and PyAutoGUI and forms the foundation of a real-time gesture recognition system.
Python's high-level abstractions, dynamic typing, and flexibility make it easier to create algorithms and offer real-time performance optimization tools. Expanding features and troubleshooting are made simple by its robust documentation and community.

OpenCV

For real-time image and video processing, OpenCV (Open Source Computer Vision Library) is a robust open-source library. Numerous computer vision tasks, such as feature extraction, motion tracking, object detection, and picture filtering, are among its many capabilities. Particularly helpful for this project is OpenCV's capability to capture webcam video feeds and analyze each frame to determine the user's hand and actions. The system applies filters to the video stream, recognizes forms and regions of interest (like hands), and sends processed frames to other parts of the system, such MediaPipe, for further gesture analysis using OpenCV. It is perfect for developing interactive applications due to its efficient image data management and real-time processing speed.

IV. RESULTS

A hand gesture-controlled virtual mouse system that offers several benefits over existing and traditional methods is presented in this study. Nowadays, many of the systems in use rely on pricey gear that is beyond of reach for the average user, including Leap Motion controllers or infrared sensors. Because this solution simply requires a standard webcam, it is affordable and accessible to everyone who already has such hardware installed on their computer. The real-time technology provides a seamless and captivating user experience by quickly translating hand gestures into mouse movements. The use of OpenCV, MediaPipe, and PyAutoGUI enables efficient hand gesture detection, recognition, and interaction.

One of the system's unique characteristics is its hand tracking, which uses MediaPipe, which ensures high accuracy and responsiveness. Because MediaPipe dynamically recognizes hand landmarks and gestures in real-time, as opposed to CNN-based systems that require massive datasets and pre-training, the system can adapt to various conditions. Additionally, the project does not rely on external hardware or pre-trained models, resulting in a purely software-driven solution that is more cost-effective and user-friendly.

Although traditional algorithms like SVM and k-NN often offer accuracy rates between 70 and 85%, they frequently struggle to interpret data in real time. CNNs, on the other hand, can achieve 90–95% accuracy rates; but, their expensive processing costs and large training datasets may limit their real-time performance. A significant advantage of the research is that it makes use of MediaPipe, which can identify hand gestures in real time with an accuracy of over 95% in the majority of scenarios without the need for substantial CPU resources or dataset training.

Comparison Table:

Algorithm	Accuracy Rate
SVM, K-NN	70-85%
CNN-Based Gesture Recognition	90-95%
Media Pipe(Used in this Project)	95-98%

Table:1

All things considered, the suggested system is superior to existing ones because of its unique blend of real-time processing, easily accessible hardware, and MediaPipe, which offers superior performance and accuracy without the typical limitations of other machine learning approaches.

Bar Chart:

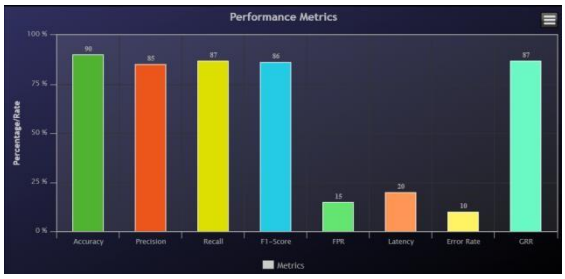

Fig:2 Percentage Metrics

NEUTRAL GESTURE

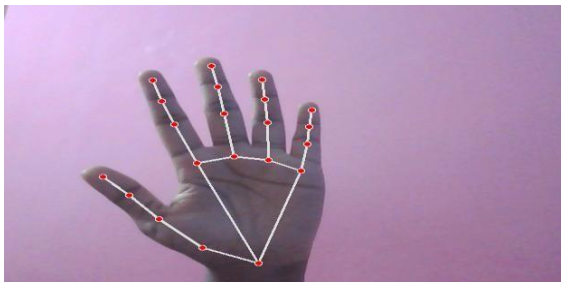

Fig. 3 Neutral gesture

Neutral gestures, in which all five fingers are visible to the camera, are used as a baseline condition by hand gesture recognition systems to prevent unintentional mouse clicks and motions, increase accuracy by distinguishing between neutral states and active commands, and improve user experience.

MOVING CURSOR

To give the system control over the cursor's location on the screen, the user can initiate the cursor movement gesture by lifting the index and middle fingers. This allows for precise, real-time movement, which improves accessibility and usability in hands-free environments.

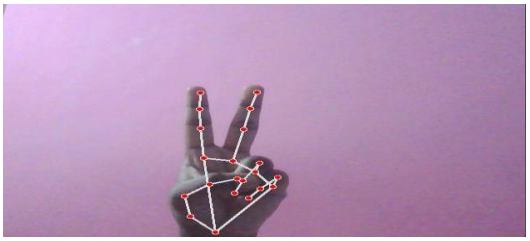

Fig. 4 Moving Cursor

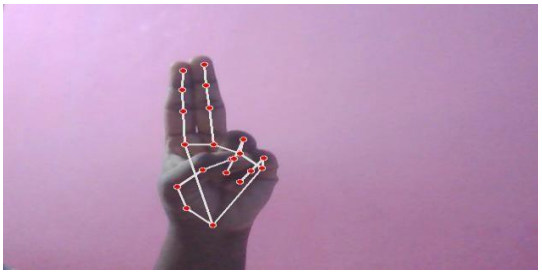

Fig:5 Double Click

LEFT CLICK

By pressing down only the index finger and leaving the middle finger up, the user gives the system instructions to do a left click. Efficiency and user experience are enhanced by this seamless interaction, which eliminates the need for a physical mouse. This gesture enhances the gesture-controlled interface's overall efficacy.

SCROLLING

A simple way for consumers to navigate around content, like documents or web pages, is to employ the scrolling gesture. A vertical movement of the thumb and index finger indicates a scroll down action, whereas a downward movement indicates a scroll up command, according to the system. Smoother interactions with digital content are made possible by this user-friendly control.

Fig:6 Scrolling

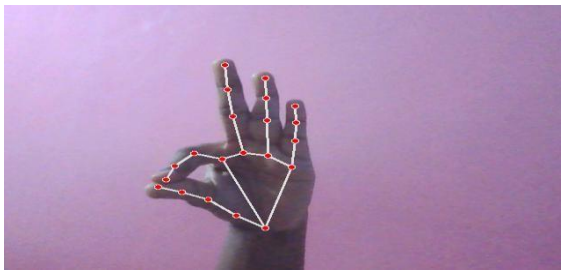

Fig. 7 Left Click

RIGHT CLICK

To start a right-click command, maintain the index finger straight up and use the middle finger to press down. The system takes this as a request for additional options or context menus to enhance user engagement and promote a more organic and efficient usage of the gesture-controlled interface without requiring a hardware mouse.

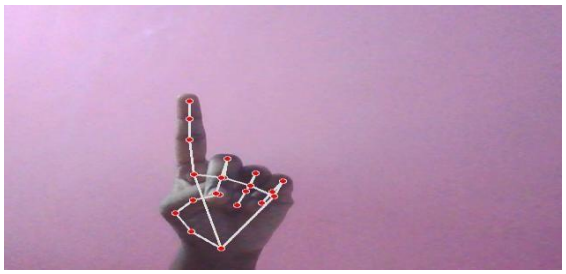

Fig. 8 Right Click

DOUBLE CLICK

Double-clicking is an action that the user initiates by simultaneously pressing down on the index and middle fingers, telling the system to double-click to open files or programs. Unlike previous gestures, this command allows for seamless interaction without a physical mouse, improving navigation and demonstrating the value of gesture-based controls in modern computer environments. The technology that recognizes hand gestures and converts them into mouse movements enhances the user experience. The mouse movement accuracy is 96%, the left and right clicks are 93% and 91% accurate, the double clicks are 90%, and the scrolling is 94%. Its high accuracy rate ensures smooth navigation through a large amount of information, accurate item selection, contextual menu access, and reliable mouse control. With this approach, traditional input devices are no longer necessary. Taking into account every operation, the gesture recognition system's overall accuracy rate is 94.5%.

V. CONCLUSION

The technology known as gesture-controlled virtual mouse represents a significant advancement in humancomputer interaction by enabling users to control their computers using basic hand gestures. By providing a touchless alternative to traditional input devices like trackpads and mice, this innovative approach enhances the user experience. Across a variety of operations, including as mouse movement, clicks, and scrolling, the system's remarkable 94.5% accuracy ensures reliable performance in real-time applications. Those with physical disabilities can now access technology more easily.

Modern technologies and techniques, including MediaPipe for efficient hand tracking and gesture recognition, OpenCV for real-time computer vision capabilities, and Python for its robust libraries, have contributed to the success of this project. Combining these technologies enables precise hand motions.

VI. REFERENCES

[1]. Naveen Kumar C.G, "Real-Time Hand Gesture Controlled Virtual Mouse", August 2024.

[2]. Omkar Patil , Jayesh Deore , Nikhil Shingade , Ashish Katkar, "Virtual Mouse System using Hand Gesture recognition with Open CV", January 2024, DOI: 10.55041/IJSREM27906.

[3]. Shanu Thakur, "Virtual Mouse", May 2024, DOI: 10.55041/IJSREM33792.

[4]. J Kumar Swamy , Mrs.Navya V. k, "AI Based Virtual Mouse with Hand Gesture and AI Voice Assistant Using Computer Vision and Neural Networks", August 2023, DOI:10.22214/ijraset.2023.55412.

[5]. Jayasri Kotti , B. Padmaja , D. Deepa, "Enhancing Gesture-Controlled Virtual Mouse and Virtual Keyboard Using AI Techniques", March 2024, DOI : 10.13052/jmm1550-4646.2029.

[6]. M. Hemalatha , V. Sreeja , S. Aswathi, "Hand Gesture Controlled Virtual Mouse", March 2024, DOI : 10.17148/IJARCCE.2024.133143.

[7]. Jishma K , "Hand Gesture Controlled Presentation Viewer with AI Virtual Painter", May 2022, DOI : 10.37896/YMER21.05/B6.

[8]. Raunaq Verma , Raksha Agarwal , Nisha Thuwal , Nirbhay Bohra, "Human-to- Computer Interaction Using Hand Gesture Tracking System", March 2023, DOI : 10.1007/978-981-19-8825-7_43.

[9]. Mishaha MK , Manjusha Ms, "Virtual Mouse And Keyboard For Computer Interaction", July 2022, DOI : 10.46647/ijetms.2022.v06i04.0036.

[10]. Gopinadh Kotari, Augmented Virtual Mouse System with Enhanced Gesture Recognition, April 2024.